Sports Medicine
https://doi.org/10.1007/s40279-025-02378-0

CURRENT OPINION

The Risks of Misunderstanding the Term "Risk Factor": A Primer with Suggestions to Improve Sports Medicine

Steven D. Stovitz[1] · Franco M. Impellizzeri[2] · Ian Shrier[3]

Accepted: 29 November 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract

The term "risk factor" is commonly used in research. Although many interpret the term to imply that the risk factor causes the outcome, others use the term to mean a marker for the outcome, which may or may not be a cause of the outcome. How the term risk factor is interpreted can importantly influence the way that study findings are applied in real world settings. For example, if a risk factor is wrongly interpreted to be a cause of an outcome when it is merely associated with the outcome because of noncausal reasons, then wasteful interventions may be developed, recommended, and implemented. The primary aims of this article are (1) to describe how varying definitions of the term risk factor can cause misunderstandings and potentially negatively impact the field of sports medicine, and (2) to propose new, more specific, terminology. We first review some basic concepts on how variables can be associated due to either causal or noncausal reasons and then discuss possible explanations for why the term risk factor continues to be misunderstood. We illustrate how using the term risk factor without further specification creates misunderstandings that can lead to the development and implementation of ineffective interventions. Finally, with the hope of improving communication and avoiding ambiguity in sports medicine, we suggest using "causal risk factor" if the evidence supports causality, "noncausal risk factor" if the evidence does not support causality, and "risk marker" for those not wishing to commit to a causal or noncausal claim.

1 Introduction

"Risk factor" is a commonly used term in research and clinical discussions, but it is an imprecise term that is interpreted differently by different authors and readers [1]. Many interpret the term to imply that the risk factor causes the outcome. However, authors often use "risk factor" less specifically to signify something that is a marker for the outcome but may not be a cause of the outcome. Different meanings of the term risk factor can have important consequences and implications in the way that study findings are interpreted and applied. For example, if a risk factor is wrongly interpreted to be a cause of an outcome when it is

Key Points

In sports medicine, the term "risk factor" is currently used with different meanings: some use it to imply causation, whereas others use it to suggest mere association without the implication for causation. These different interpretations lead to different recommendations.

We recommend against using the term "risk factor" without clarification to avoid misunderstanding. Specifically, if changing the factor is expected to change the outcome, then use the term "causal risk factor." If changing the factor is not expected to change the outcome, then use "noncausal risk factor." For those not wishing to commit to a causal or noncausal claim, we suggest the more generic and agnostic term "risk marker."

Sports medicine academicians and clinicians who write scientific and lay articles would be served well by recognizing the varying definitions and then being precise with their language to avoid misinterpretations.

✉ Franco M. Impellizzeri
Franco.Impellizzeri@uts.edu.au

1 Department of Family Medicine and Community Health, University of Minnesota, Minneapolis, MN, USA

2 Faculty of Health, Human Performance Research Centre, University of Technology Sydney, Sydney, NSW 2007, Australia

3 Centre for Clinical Epidemiology, Lady Davis Institute, Jewish General Hospital, McGill University, Montreal, Canada

Published online: 19 December 2025

merely associated with the outcome because of noncausal reasons, then wasteful interventions may be developed, recommended, and implemented.

In the hope of improving the research and practice of sports medicine, the primary aim of this article is to describe how varying definitions of the term risk factor can cause misunderstandings and negatively impact the field of sports medicine. We first review some basics regarding how variables can be associated due to either causal or noncausal reasons and then discuss reasons why the term risk factor continues to be misunderstood. We illustrate how using the term risk factor without further specification creates misunderstandings that can lead to the development and implementation of ineffective interventions. Finally, we aim to improve sports medicine science and communication between researchers and practitioners by offering suggestions for terminology to clarify academic discussions.

2 Causal Versus Noncausal Associations

Central to a discussion of the term risk factor is the need to recognize that variables may be associated due to either causal or noncausal mechanisms [2]. Imagine two hypothetical variables, *X* and *Y*. These variables are associated if knowledge about *X* provides information about *Y*, or vice versa. Figure 1 is a schematic of four main ways in which *X* and *Y* can be associated. Causal associations occur when *X* causes *Y* (Fig. 1a) or *Y* causes *X* (Fig. 1b). Alternatively, noncausal associations can occur as in Fig. 1c or Fig. 1d. Figure 1c demonstrates confounding bias. There are many definitions of confounders and confounding. We are using a structural approach where confounding bias occurs if there are additional uncontrolled variables that are causes of both *X* and *Y* (e.g., Fig. 1c). Figure 1d shows a form of selection bias where both *X* and *Y* cause another variable and that variable is controlled in some way, e.g., by the design restricting the variable or the analysis controlling for the variable. Recognizing how variables can be associated through noncausal mechanisms, such as confounding and selection bias, is necessary to understand how associations, and thus the term risk factor, can be misunderstood. Our description of these noncausal pathways is incomplete but can serve as a necessary starting point to discuss the differing definitions of risk factors. For more complete and in-depth discussions about the structural approach to biases, Hernán et al. [3], Greenland [4], Hernán [5] and Kenah [6].

3 Reasons for Misunderstanding

3.1 Different Definitions of the Term "Risk Factor"

In the year 2001, the 4th edition of "A Dictionary of Epidemiology", [7] acknowledged that *"the term risk factor is rather loosely used, with any of the following meanings: (1) An attribute or exposure that is associated with an increased probability of a specified outcome, such as the occurrence of a disease. Not necessarily a causal factor (a risk marker), (2) An attribute or exposure that increases the probability of occurrence of disease or other specified outcome (a determinant), or (3) A determinant that can be modified by intervention, thereby reducing the probability*

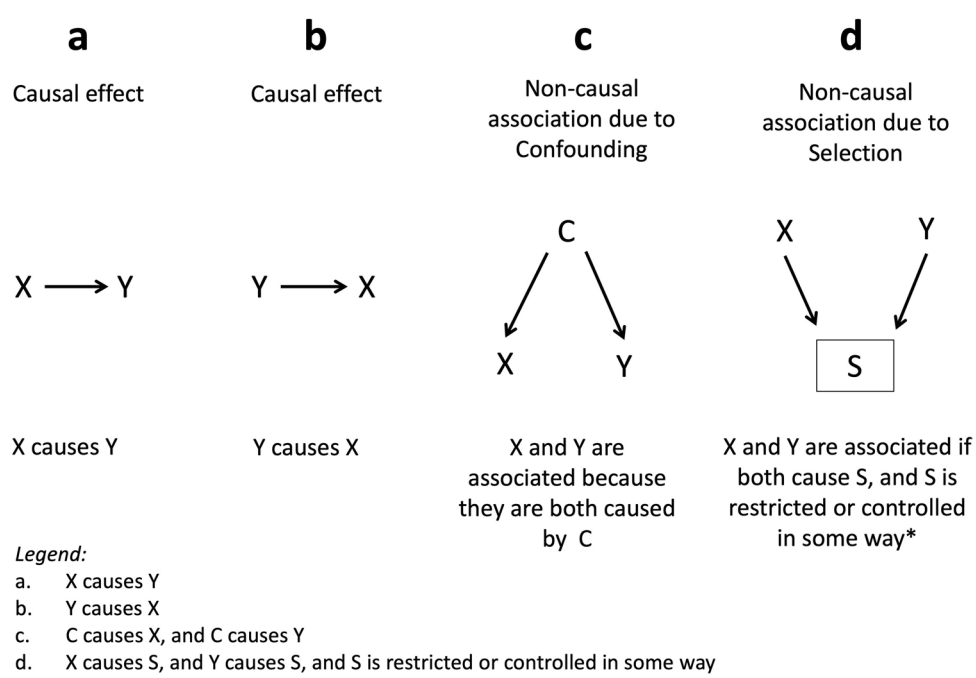

Fig. 1 Mechanisms by which variables *X* and *Y* can be associated. *The box around S signifies that the value is known. This may be through restricting the study to only those with one level of S or statistical control of S. Adapted with permission from Stovitz et al. [2]

of occurrence of disease or other specified outcomes. To avoid confusion, it may be referred to as a modifiable risk factor." [7]

In 2014, a later edition of the textbook [8] attempted to minimize confusion about the term risk factor by defining it as a variable causally related to the outcome, and then using "risk marker" as a variable not causally related to the outcome [8]. Risk factor and risk marker together were grouped under the umbrella term "risk indicators." [8] However, these definitions have been criticized [9] and not widely adopted, with many still using the term risk factor loosely to signify any associated variable, whether causal or noncausal. In 2016, Huitfeldt recommended avoiding the term risk factor [10]. Instead, he suggested that investigators specify their intent (e.g., diagnostic factor, a prognostic factor, an etiological factor, or a treatment effect) within the context of the particular research question [10]. Given the lack of consensus about the precise definition in the discipline of epidemiology, it is understandable that there is confusion in the discipline of sports medicine.

Confusion about the term risk factor likely begins with confusion about the term risk. In epidemiology, the term risk does not signify causality [11]. The risk of having a heart attack while golfing is simply the number of people who suffer a heart attack while golfing divided by the number of people golfing. It does not necessarily imply causality, and as such, the term risk in this context does not provide information about the best course of action to take in the future. For example, having had knee anterior cruciate ligament (ACL) reconstructive surgery is a risk for a new ACL tear. That does not mean that the surgery causes the ACL tear or that an ACL tear would be prevented by not having had the surgery. In epidemiology, the term risk is used without expectation that the outcome would increase or decrease because of the risk being present or absent. Contrary to the above definition from epidemiology, many use the word risk as if it has causal implications [12] such as in the relationships shown in Fig. 1a or Fig. 1b. Furthermore, the term risk may be interpreted similar to the lay term risky. "Risky" is defined as "dangerous; hazardous; fraught with risk." [13] This connotes that the variable represents an action that has a causal effect and is something that is to be avoided, and thus is not consistent with the definition of the term risk in epidemiology.

3.2 Associations Between Variables have Different Clinical Interpretations

In epidemiology and statistics, risk factors refer to associations between variables. These associations can be *descriptive*, reporting the distribution of variables, *predictive*, estimating the probability of concurrent (e.g., diagnosis) or future events (e.g., prognosis) with a model that has undergone some validation, or *causal*, changing what has or will happen [14, 15]. How a risk factor is interpreted affects how it can be used in research and the effects that it should have (or not have) on clinical practice. A noncausal risk factor, i.e., one that is merely associated with the outcome of interest, can be used to describe data, to generate hypotheses, or as a variable in a prediction model (for prognosis or diagnosis) [16]. When a risk factor is used in a prediction model or is referred to as a "predictor," it simply means that it is a feature included in the model, typically multivariable, and does not imply that the risk factor causes the outcome. In addition, claims that a risk factor can be used to predict future outcomes requires evidence of its predictive performance, including calibration, discrimination, and both internal and external validation, rather than merely the association with the outcome [17]. When evidence for the variable being a risk factor is derived from descriptive or predictive analyses, then the risk factor should not be considered causal and should not serve as a target for intervention [15]. When evidence suggests that the risk factor caused the outcome, i.e., when alternative biasing pathways to the outcome are eliminated through research design or statistical analyses, then the risk factor can appropriately be considered a target for intervention.

3.3 Ambiguity About the Word "Modifiable" When Describing Risk Factors

The confusion about risk factors becomes even more salient when people use the term "modifiable risk factors." The word modifiable only signifies that we can intervene to change the value of the risk factor. It does not necessarily mean that changing the risk factor changes the outcome. While the use of "modifiable risk factor" for factors not causally related to the outcome is technically correct, it is evident that this can cause confusion and lead readers (voluntarily or involuntarily) to interpret these factors as causal. As an example of this point, a very recent systematic review titled "Modifiable risk factors for lower-extremity injury…" used a definition for risk factor of "any factor associated with a lower-extremity injury." [18] There was no specific evaluation of the strength of evidence for causation for any of the risk factors identified (e.g., using Risk Of Bias In Non-randomized Studies (ROBINS)-of Exposure (E)/of Interventions (I)) [19, 20]. We agree with the authors' conclusions that the observed associations in their review should serve "as a foundation for investment and action to reduce the burden… [of] lower-extremity injuries." [18] Working off this foundation should be done through future studies evaluating causal effects. However, we have concern that many readers would skip the call for future studies into causal effects, thinking

that the associations support intervening on the modifiable risk factors despite the lack of evidence that the risk factors caused the outcomes.

3.4 Causal Illusion

Unfortunately, even the most specific language may not prevent miscommunication because the human mind has evolved to perceive causality from coincidence or from events that occur sequentially (*post hoc ergo propter hoc* fallacy) [21]. A study by Haber et al. [22] found that researchers tend to interpret apparently neutral terms such as "association" with some causal implication. Furthermore, context affects people's belief that events are related due to causality [23]. Consider the observational study of motorcycle riders that found a higher risk for serious injuries in those who rode without a helmet than in those who rode with a helmet [24]. We believe that riding a motorcycle without a helmet is dangerous, and so we are susceptible to believe that the action of not wearing a helmet is a causal risk factor for a serious injury. However, the same study also found a higher risk for serious injuries in riders without health insurance compared with riders with health insurance [24]. It is implausible that not having health insurance causes injuries. Rather, a noncausal association is expected because risk takers may be more likely to (1) take a risk of not having insurance and (2) ride without a helmet. This example illustrates how our preconceived notions can affect our interpretations. D'Amico et al summarized 14 instances where the medical community unknowingly implemented harmful changes due to wrongly interpreting noncausal associations in observational studies as evidence of causality. [25]

3.5 Summary of Reasons for Misunderstandings

In summary, it is understandable that the term risk factor is often misunderstood in sports medicine. First, it has not been well defined, even in the field of epidemiology. Also, studies often do not clearly identify whether their projects are geared toward descriptive, predictive, or explanatory goals. The word "modifiable" is used when the variable that is the risk factor can be modified regardless of whether it has a causal relationship with the outcome, i.e., regardless of whether changing the risk factor will change the outcome. This can easily confuse readers and is why, as discussed later, we suggest not using the term "modifiable" unless the risk factor has a causal effect. Finally, people seem to naturally perceive causality even when there is no evidence to suggest that there is a causal relationship between the risk factor variable and the outcome.

4 Miscommunication in Sports Medicine

Researchers in various medical disciplines imply causality despite analyzing observational data in a manner directed at finding associations that may be due to noncausal reasons [26]. Sports medicine is no exception, often using the term risk factor without specification and then using causal language in its interpretation. For example, a recent study on risk factors for shoulder injuries in swimmers used the "strength" of the evidence as determined in a previous systematic review [27] based on the significance, direction, and magnitude of the associations, without determination that the associations were due to evidence of causality. The authors then asked expert swimming practitioners "based on [their] experience, what specific risk factors do [they] believe are contributing to shoulder injury in swimmers." [28] If you "contribute" to something, you change it, i.e., the use of the term "contributing" implies causation. The participants were also asked to rate the "modifiability," which, again, implies that the modification of these factors can have an effect on injury risk. In summary, the review evaluated studies of risk factors without a determination of whether the associations were due to causal or noncausal reasons but then used the results as if the risk factors caused the outcomes.

We agree that interventions can be justified even when empirical evidence of causality is lacking and understand that this is often done in sports medicine where there is a paucity of research evaluating causal effects. Our point is that the rationale for using noncausal evidence to support causal claims should be communicated clearly. This decision should be predicated on a sound theoretical rationale explaining why the risk factor likely has a beneficial effect, while also considering the costs, risk of harm, and other negative consequences. We recognize that it is ultimately the practitioners and policymakers who will decide whether to intervene based on their particular situation.

Another recent systematic review and meta-analysis "identified and analyzed the risk factors of lower extremity injuries in elite women football players to improve career and health outcomes." [29] The authors found eight predictors among four studies "that influence the risk of lower extremity injuries." These four studies examined only associations without providing evidence that these factors are causal. However, the authors of this systematic review and meta-analysis still concluded that the risk of lower extremity injuries can be decreased by focusing on their physical characteristics, knee postures, and movements [29]. Thus, they took predictive evidence (which is not specifically causal) and interpreted it as explanatory

(i.e., as if the evidence were causal). Conflating predictive and causal research (see Sect. 3.2) is a common source of confusion in the literature and is reflected in how risk factors are interpreted.

We mention these studies as symbolic examples of a broader issue in sports medicine, and they should not be interpreted as personal criticism of the authors of these studies. Similar examples can be found in many medical disciplines, including epidemiology. [26]

5 Conclusion

The term risk factor is fraught with misunderstandings that can have potential practical consequences such as wasting resources on poorly identified targets for interventions. Some researchers define the term to signify factors that are merely associated with outcomes regardless of their causal effects, and others define the term to signify causal factors. These varying definitions are understandable as there has not been a strong consensus within the scientific community. Information that lacks knowledge about causal effects can be used for hypothesis generation, e.g., to ask: Is the factor a cause? However, this hypothesis needs to be examined further before making causal claims or recommending a change in the risk factor. If practitioners still want to intervene on these risk factors when the evidence on causality is lacking, the rationale for this decision should be explained and there should be consideration of the costs, risk of harm, or other negative consequences.

5.1 Suggestions for Improvement

To improve the research and practice of sports medicine, we suggest the following:

1. Use "causal risk factor" if the evidence supports causality, and use "noncausal risk factor" if the evidence does *not* support causality. For those not wishing to commit to a causal or noncausal claim, we suggest the term "risk marker," which we hope can be understood as being agnostic toward causal claims.
   - The above recommendation coincides with the usual English definition of the word "marker" and is consistent with the 4th edition of A Dictionary of Epidemiology published in 2001 [7].
   - It asks authors to contemplate whether their results warrant causal claims.

     Determining when evidence supports causality is a controversial topic and beyond the scope of this article. Discussions of this topic can be found in Vandenbroucke et al. [30] and Dahabreh and Bibbins-Domingo [31].
2. If not following #1, at minimum define your definition of the term risk factor.
   - If authors prefer a different definition, they should define risk factors as causal, noncausal, or agnostic to causality the first time they are used and then be consistent in using that definition.
3. Journals can be a stopgap
   - Since it is impractical to believe that all authors would spontaneously follow the above suggestions, another option would be Huitfeldt's plea that journals serve as a stopgap [10]. Huitfeldt has suggested a taboo on the term risk factor and to force authors to specify whether the variable is "a diagnostic factor, a prognostic factor, an aetiological factor, or a treatment effect." [10]. His categorization simply replaces "causal risk factor" with "etiological risk factor" or "treatment effect" depending on the context. Regardless of the terminology used, we therefore suggest that journal editors be proactive in preventing misinterpretation by requiring authors to clarify their intended meaning when using the term risk factor (or similar expressions).
4. Use the term "modifiable" as an adjective to risk factor *only* when referring to causal risk factors
   - Due to the high likelihood of misunderstanding, references to modifiability should be used only in relation to causal risk factors. The fact that a factor can be modified but is not causally related to the outcome is irrelevant and may mislead the reader.
5. Education is needed
   - A large part of the solution requires teaching principles of causality to everyone in the health sciences. The educational content should include recognition that people are easily fooled.
   - Without both an understanding of the principles of causality and a recognition that relationships between variables are often misunderstood, sports medicine clinicians and researchers will continue to be prone to misunderstandings about factors that should be acted upon and those that are merely associated with outcomes owing to noncausal reasons. These misunderstandings can lead to either recommending against effective treatment or recommending ineffective treatment.

Funding No external sources of funding were used in the preparation of this article.

Declarations

Conflicts of interest The authors have no potential financial conflicts of interest with the content of this article. S.D.S. and F.M.I. are employed by universities. S.D.S. is primarily employed as a physician and educator, but grants and publications are part of his performance evaluation. S.D.S. is an associate editor for *British Journal of Sports Medicine* and *Medicine and Science in Sport and Exercise*. A primary academic responsibility of F.M.I. is to obtain grants and publish papers, including one current grant received for a meta-research related topic (not financially related to the current article). F.M.I. is Editor-in-Chief of *Science and Medicine in Football*, an editorial board member of the *Journal of Science and Medicine in Sport* and *Scandinavian Journal of Medicine & Science in Sports*, and a member of the managing board of the *Peer Community In Human & Movement Sciences*. I.S. is retired but continues to hold grants and conduct research. I.S. is on the editorial board of *Journal of Science and Medicine in Sport*, *Scandinavian Journal of Medicine & Science Sports*, and *Translational Sports Medicine*.

Ethics and Consent No patient data were used, and Institutional Review Board approval was not applicable.

Author Contributions All authors participated equally in the article, including the design, intellectual content, writing, and editing. All authors read and approved the final version of the manuscript.

References

1. Schooling CM, Jones HE. Clarifying questions about "risk factors": predictors versus explanation. Emerg Themes Epidemiol. 2018;15:10. https://doi.org/10.1186/s12982-018-0080-z.
2. Stovitz SD, Verhagen E, Shrier I. Distinguishing between causal and non-causal associations: implications for sports medicine clinicians. Br J Sports Med. 2019;53(7):398–9. https://doi.org/10.1136/bjsports-2017-098520.
3. Hernán MA, Hernández-Díaz S, Robins JM. A structural approach to selection bias. Epidemiology. 2004;15(5):615–25. https://doi.org/10.1097/01.ede.0000135174.63482.43.
4. Greenland S. Response and follow-up bias in cohort studies. Am J Epidemiol. 1977;106(3):184–7. https://doi.org/10.1093/oxfordjournals.aje.a112451.
5. Hernán MA. Invited commentary: selection bias without colliders. Am J Epidemiol. 2017;185(11):1048–50. https://doi.org/10.1093/aje/kwx077.
6. Kenah E. A potential outcomes approach to selection bias. Epidemiology. 2023;34(6):865–72. https://doi.org/10.1097/EDE.0000000000001660.
7. Last JM, International Epidemiological Association. A Dictionary of epidemiology. 4th ed. Oxford University Press; 2001.
8. Porta M. A dictionary of epidemiology. In: A dictionary of epidemiology. Oxford University Press; 2014. https://doi.org/10.1093/acref/9780199976720.001.0001/acref-9780199976720. Accessed May 31, 2024.
9. Bhopal RS. Concepts of epidemiology: integrating the ideas, theories, principles, and methods of epidemiology. 3rd ed. Oxford University Press; 2016.
10. Huitfeldt A. Is caviar a risk factor for being a millionaire? BMJ Online. 2016. https://doi.org/10.1136/bmj.i6536.
11. Benichou J, Palta M. Rates, risks, measures of association and impact. In: Ahrens W, Pigeot I, editors. Handbook of epidemiology. Springer; 2005. p. 89–156. https://doi.org/10.1007/978-3-540-26577-1_3.
12. Risk noun - Definition, pictures, pronunciation and usage notes | Oxford Advanced Learner's Dictionary at OxfordLearnersDictionaries.com. https://www.oxfordlearnersdictionaries.com/definition/english/risk_1. Accessed 12 Jan 2025.
13. Risky, adj. meanings, etymology and more | Oxford English Dictionary. https://www.oed.com/dictionary/risky_adj. Accessed 31 May 2024.
14. Losciale JM, Bullock GS, Collins GS, et al. Description, prediction, and causation in sport and exercise medicine research: resolving the confusion to improve research quality and patient outcomes. J Orthop Sports Phys Ther. 2023. https://doi.org/10.2519/jospt.2023.11773.
15. Hernán MA, Hsu J, Healy B. A second chance to get causal inference right: a classification of data science tasks. Chance. 2019;32(1):42–9. https://doi.org/10.1080/09332480.2019.1579578.
16. Riley RD, Hayden JA, Steyerberg EW, et al. Prognosis Research Strategy (PROGRESS) 2: prognostic factor research. PLoS Med. 2013;10(2):e1001380. https://doi.org/10.1371/journal.pmed.1001380.
17. Collins GS, Dhiman P, Ma J, et al. Evaluation of clinical prediction models (part 1): from development to external validation. BMJ. 2024;384:e074819. https://doi.org/10.1136/bmj-2023-074819.
18. Whittaker JL, Räisänen AM, Martin C, et al. Modifiable risk factors for lower-extremity injury: a systematic review and meta-analysis for the Female, Woman and Girl Athlete Injury Prevention (FAIR) consensus. Br J Sports Med. 2025. https://doi.org/10.1136/bjsports-2025-109902.
19. Higgins JPT, Morgan RL, Rooney AA, et al. A tool to assess risk of bias in non-randomized follow-up studies of exposure effects (ROBINS-E). Environ Int. 2024;186:108602. https://doi.org/10.1016/j.envint.2024.108602.
20. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016;355:i4919. https://doi.org/10.1136/bmj.i4919.
21. Matute H, Blanco F, Yarritu I, Díaz-Lago M, Vadillo MA, Barberia I. Illusions of causality: how they bias our everyday thinking and how they could be reduced. Front Psychol. 2015;6:888. https://doi.org/10.3389/fpsyg.2015.00888.
22. Haber NA, Wieten SE, Rohrer JM, et al. Causal and associational language in observational health research: a systematic evaluation. Am J Epidemiol. 2022;191(12):2084–97. https://doi.org/10.1093/aje/kwac137.
23. Fugelsang JA, Thompson VA. Strategy selection in causal reasoning: when beliefs and covariation collide. Can J Exp Psychol/Revue canadienne de psychologie expérimentale. 2000;54(1):15–32. https://doi.org/10.1037/h0087327.
24. Brown CVR, Hejl K, Bui E, Tips G, Coopwood B. Risk factors for riding and crashing a motorcycle unhelmeted. J Emerg Med. 2011;41(4):441–6. https://doi.org/10.1016/j.jemermed.2009.07.024.
25. D'Amico F, Marmiere M, Fonti M, Battaglia M, Belletti A. Association does not mean causation, when observational data were misinterpreted as causal: the observational interpretation fallacy. J Eval Clin Pract. 2025;31(1):e14288. https://doi.org/10.1111/jep.14288.

26. Ramspek CL, Steyerberg EW, Riley RD, et al. Prediction or causality? A scoping review of their conflation within current observational research. Eur J Epidemiol. 2021;36(9):889–98. https://doi.org/10.1007/s10654-021-00794-w.
27. McKenzie A, Larequi SA, Hams A, Headrick J, Whiteley R, Duhig S. Shoulder pain and injury risk factors in competitive swimmers: a systematic review. Scand J Med Sci Sports. 2023;33(12):2396–412. https://doi.org/10.1111/sms.14454.
28. McKenzie AK, Hams A, Headrick J, et al. Identifying evidence-practice gaps for shoulder injury risk factors in competitive swimmers: uniting literature and expert opinion. Br J Sports Med. 2024. https://doi.org/10.1136/bjsports-2023-108023.
29. Gashi F, Kovacic T, Gashi AI, Boshnjaku A, Shalaj I. Predicting risk factors of lower extremity injuries in elite women's football: systematic review and meta-analysis. Sports. 2023;11(9):187. https://doi.org/10.3390/sports11090187.
30. Vandenbroucke JP, Broadbent A, Pearce N. Causality and causal inference in epidemiology: the need for a pluralistic approach. Int J Epidemiol. 2016;45(6):1776–86. https://doi.org/10.1093/ije/dyv341.
31. Dahabreh IJ, Bibbins-Domingo K. Causal inference about the effects of interventions from observational studies in medical journals. JAMA. 2024;331(21):1845–53. https://doi.org/10.1001/jama.2024.7741.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.